

Use-Case Flow

UC-05 — Review Inspection & Sign Off
Second-Hand Product Escrow Marketplace
Dhanush R · PES1UG24AM427 · Problem #39 · SE Lab 1

USE-CASE OVERVIEW

Use Case	UC-05 — Review Inspection & Sign Off
Actors	Buyer · Second-Hand Product Escrow Marketplace (System)
Goal	The buyer reviews the Inspection Agent's report and either accepts the product condition (triggering escrow release eligibility) or raises a dispute (freezing escrowed funds).

PRECONDITIONS

- Buyer has an active order for a second-hand product.
- Buyer's payment is locked in escrow.
- Order status is exactly "Under Inspection".
- Inspection Agent has submitted a physical inspection report for the order.

POSTCONDITIONS

- If buyer accepts: buyer sign-off is recorded and order status transitions to "Awaiting Release", making the order eligible for escrow funds release.
- If buyer raises a dispute: a dispute ticket is created, escrow funds are frozen, and the order is routed for investigation; escrow release is blocked until the dispute is resolved.

MAIN SUCCESS SCENARIO

1. Buyer opens the active order awaiting inspection confirmation.
2. System displays the Inspection Agent's submitted inspection report including condition rating, description text, and any attached photos.
3. Buyer reviews the inspection result.
4. Buyer selects the option to accept the inspected product.
5. System records the buyer's authenticated sign-off (user ID + UTC timestamp) against the order.
6. System verifies that both inspection_verified = true AND buyer_signed_off = true.
7. System transitions the order status to "Awaiting Release", enabling escrow fund release to the seller.
8. Use case ends successfully.

ALTERNATE FLOW — BUYER DISPUTES INSPECTION

- 4A1. Buyer determines that the product condition is not acceptable.**
- 4A2. Buyer selects "Raise Dispute" instead of accepting the product.**

4A3. System creates a dispute ticket with a unique ID, the buyer's user ID, the order ID, a UTC timestamp, and status "Open"; escrowed funds are frozen within 5 seconds.

4A4. System records the dispute against the order and routes it to the Inspection Agent queue for investigation.

4A5. The transaction does not proceed to escrow release. Any release or withdrawal request for the order returns error code FUNDS_FROZEN until the dispute is resolved.

EXCEPTIONS & NOTES

- If the order status is not "Under Inspection" when this use case is invoked, the system returns error code INVALID_ORDER_STATE and the use case terminates.
- If the buyer's authentication session has expired, the system prompts re-authentication before recording any sign-off or dispute action.
- Only one sign-off or dispute action is permitted per order; duplicate submissions are rejected.